

Navier–Stokes Existence or Breakdown

Notes from a lecture by Javier Gómez-Serrano

(These notes started as an AI-cleaned transcript of the talk.)

1. Introduction

This lecture concerns the Navier–Stokes existence or breakdown problem, one of the seven Millennium Prize Problems posed by the Clay Mathematics Institute around the year 2000.

The problem asks, roughly:

given smooth initial data for the 3D incompressible Navier–Stokes equations,

- do smooth solutions exist for all time?
- or can singularities form in finite time?

The talk emphasizes that, although the Clay problem remains open, the field has seen substantial activity and major conceptual progress in recent years.

2. Historical background

2.1. Leonardo da Vinci and early observations

Leonardo da Vinci studied fluid motion and apparently coined the term *turbulence* (Italian: *turbolenza*). His observations already captured a multiscale picture of fluid flow: large eddies feeding into smaller ones.

He wrote:

Observe the motion of the surface of the water, which resembles that of hair. The water has eddying motions, one part of which is due to the principal current and the other to the random and reverse motion.

So, already around 1500, there was a qualitative awareness of the cascade-like structure of fluid motion.

2.2. Bernoulli, d'Alembert, Euler, Prandtl

A few milestones:

- **Bernoulli** (Daniel Bernoulli, 1700–1782) related higher fluid velocity to lower pressure, leading to explanations such as lift on airplane wings. He published **Hydrodynamica** in 1738.
- **d'Alembert** (Jean le Rond d'Alembert, 1717–1783) considered drag and argued that an inviscid fluid should produce zero drag, in contradiction with physical observations (1752). This is the famous *d'Alembert paradox*.
- **Euler** (Leonhard Euler, 1707–1783) wrote the incompressible Euler equations (1757), describing ideal fluids without viscosity.
- **Prandtl** (Ludwig Prandtl, 1875–1953) explained the paradox through *boundary layer theory* (1904): even very small viscosity can produce substantial drag near boundaries.

Thus the mathematical theory of fluid mechanics emerged from a combination of physical observation and formal modeling.

3. The Euler and Navier–Stokes equations

Let $u = u(x, t)$ denote the velocity field, $p = p(x, t)$ the pressure, ρ the density, ν the viscosity, and $f = f(x, t)$ an external force.

The incompressible fluid equations are

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f$$

together with the incompressibility condition

$$\nabla \cdot u = 0.$$

- If $\nu = 0$, one obtains the incompressible Euler equations.
- If $\nu > 0$, one obtains the incompressible Navier–Stokes equations.

These equations express Newton’s law $F = ma$ for a fluid parcel.

3.1. The equations

The incompressible Navier–Stokes equations above combine the local acceleration $\partial_t u$, the nonlinear advection $(u \cdot \nabla)u$, the pressure gradient ∇p , and the viscous diffusion $\nu \Delta u$. Together they govern the evolution of a fluid velocity field.

3.2. The nonlinear term

The main mathematical difficulty comes from the nonlinear advection term

$$(u \cdot \nabla)u.$$

This term means that the velocity field affects its own evolution. The equation is therefore nonlinear, and this raises the possibility of dramatic phenomena such as growth of gradients or formation of singularities.

3.3. Incompressibility

The condition

$$\nabla \cdot u = 0$$

means the fluid is incompressible: parcels preserve volume. This prevents simple compression-type blowup mechanisms that might occur in compressible flow.

3.4. The pressure equation

Taking the divergence of the Navier–Stokes equation and using $\nabla \cdot u = 0$ gives the Poisson equation for the pressure:

$$-\Delta p = \nabla \cdot ((u \cdot \nabla)u) = \sum_{i,j=1}^3 \partial_i \partial_j (u_i u_j).$$

Thus p is determined globally by u at each instant. Solving the Poisson equation with the fundamental solution $G(x) = \frac{1}{4\pi|x|}$ of $-\Delta$ in $\mathbb{R}^3$ gives the singular integral representation

$$p(x, t) = \frac{1}{4\pi} \int \left(\frac{3(x_i - y_i)(x_j - y_j)}{|x - y|^5} - \frac{\delta_{ij}}{|x - y|^3} \right) u_i(y, t) u_j(y, t) dy.$$

(Summation over $i, j = 1, 2, 3$ is implied.) This nonlocal coupling between u and p is a key source of mathematical difficulty: the pressure instantaneously transmits information across the entire domain.

4. The Clay Millennium problem

The 3D incompressible Navier–Stokes problem, in one standard formulation, is:

Given smooth divergence-free initial data u_0 on either $\mathbb{R}^3$ or the torus $\mathbb{T}^3$, consider the evolution

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u \quad \nabla \cdot u = 0 \quad u(\cdot, 0) = u_0.$$

The question is whether one of the following holds:

1. **Global regularity:** for every smooth initial datum, there exists a smooth solution for all time.
2. **Finite-time breakdown:** there exists smooth initial datum for which a singularity forms in finite time.

A singularity means that some norm of the solution, typically involving u or its derivatives, becomes unbounded in finite time.

5. Three broad scenarios

The speaker described three conceptual possibilities:

1. **Global smoothness and uniqueness**
Smooth solutions remain smooth for all time.
2. **Singularity formation, but uniqueness persists**
Singularities may form, yet the evolution remains uniquely determined in an appropriate sense.
3. **Singularity formation and nonuniqueness**
Singularities occur, and beyond them there may be multiple solutions.

The Clay problem is usually presented as distinguishing scenario 1 from scenarios 2 or 3 combined.

A different emphasis, associated with Ladyzhenskaya, is to distinguish uniqueness from nonuniqueness.

6. Weak solutions

To discuss global existence, one introduces weaker notions of solution.

6.1. Definition of weak solution

A **weak solution** is a function u that satisfies the Navier–Stokes equations only after testing against smooth divergence-free test functions.

For simplicity, consider the unforced equation on $\mathbb{R}^3$ or $\mathbb{T}^3$:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u \quad \nabla \cdot u = 0.$$

Then u is a weak solution if for every smooth compactly supported divergence-free test field φ ,

$$\int_0^\infty \int (u \cdot \partial_t \varphi + (u \otimes u) : \nabla \varphi + \nu \nabla u : \nabla \varphi) dx dt + \int u_0(x) \cdot \varphi(x, 0) dx = 0.$$

This is obtained by multiplying the PDE by φ and integrating by parts.

A strong solution is always a weak solution, but the converse need not hold.

6.2. Leray–Hopf weak solutions

A **Leray–Hopf weak solution** is a weak solution that also satisfies the energy inequality

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds \leq \frac{1}{2} \|u_0\|_{L^2}^2$$

for all t .

This means the kinetic energy is nonincreasing, except for dissipation through viscosity.

6.3. Leray–Hopf theorem

Leray (1934) proved the foundational existence result for 3D incompressible Navier–Stokes: for any divergence-free initial data $u_0 \in L^2(\mathbb{R}^3)$, there exists at least one global weak solution

$$u \in L_t^\infty L_x^2 \cap L_t^2 \dot{H}_x^1$$

satisfying the energy inequality for all t . Moreover, the solution is smooth except possibly on a small exceptional set; in particular, the set of singular times has zero $\frac{1}{2}$ -dimensional Hausdorff measure.¹ Hopf (1951) later extended this to bounded domains with no-slip boundary conditions.²

Thus:

- global weak solutions exist for any L^2 initial data;
- but it is unknown whether they remain smooth or unique in full generality.

If one could prove that every Leray–Hopf weak solution is actually smooth, the Clay problem would be solved in the positive direction.

7. The two-dimensional case

In two spatial dimensions, the situation is much better.

7.1. Ladyzhenskaya’s theorem

Ladyzhenskaya (1958, 1969) proved global well-posedness for 2D incompressible Navier–Stokes: for any divergence-free $u_0 \in L^2(\mathbb{R}^2)$, there exists a unique global solution

$$u \in C([0, \infty); L^2) \cap L_{\text{loc}}^2((0, \infty); H^1)$$

that is smooth for all $t > 0$ and depends continuously on the initial data.³

The key point is that certain inequalities, now often called Ladyzhenskaya inequalities, provide a stronger control in two dimensions than in three. In 2D, the inequality

$$\|u\|_{L^4} \leq C \|u\|_{L^2}^{\frac{1}{2}} \|\nabla u\|_{L^2}^{\frac{1}{2}}$$

holds, which together with the energy inequality gives enough control to close the estimates. In 3D, the analogous inequality

$$\|u\|_{L^4} \leq C \|u\|_{L^2}^{\frac{1}{4}} \|\nabla u\|_{L^2}^{\frac{3}{4}}$$

has a higher exponent on the ∇u term, and the same strategy breaks down.

So:

- in 2D Navier–Stokes, the global regularity theory is complete;
- in 3D, it remains open.

8. Partial regularity

Even if singularities exist, can we say how large the singular set is?

¹J. Leray, “Sur le mouvement d’un liquide visqueux emplissant l’espace,” **Acta Math.** 63, 193–248 (1934). DOI Zbl

²E. Hopf, “Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen,” **Math. Nachr.** 4, 213–231 (1951). DOI Zbl

³O. A. Ladyzhenskaya, “Solution ‘in the large’ of the nonstationary boundary value problem for the Navier–Stokes system in two space variables,” **Comm. Pure Appl. Math.** 12, 427–433 (1959). DOI Zbl also **The Mathematical Theory of Viscous Incompressible Flow**, 2nd ed., Gordon & Breach (1969).

8.1. Suitable weak solutions

A **suitable weak solution** is a weak solution that additionally satisfies the **local energy inequality**

$$\partial_t \left(\frac{|u|^2}{2} \right) + \operatorname{div} \left(\left(\frac{|u|^2}{2} + p \right) u \right) - \nu \Delta \left(\frac{|u|^2}{2} \right) + \nu |\nabla u|^2 \leq 0$$

in the sense of distributions. This local form of energy dissipation is a stronger condition than the global energy inequality and is essential for partial regularity arguments.

8.2. Scheffer

Scheffer (1977) proved that the singular set of a suitable weak solution has parabolic Hausdorff dimension at most $\frac{5}{3}$ (later improved to at most 2).⁴

8.3. Caffarelli–Kohn–Nirenberg

Caffarelli, Kohn, and Nirenberg (1982) improved this dramatically: the one-dimensional parabolic Hausdorff measure of the singular set is zero.⁵

Concretely, this implies that the singular set S satisfies $P^1(S) = 0$, meaning its one-dimensional parabolic Hausdorff measure vanishes. In particular, S cannot contain any curve of positive length in spacetime, so possible singularities are extremely sparse.

This partial regularity result historically supported optimism for global regularity: if singularities exist, they must be highly constrained.

9. Vorticity formulation

A central derived quantity is the **vorticity**

$$\omega = \nabla \times u.$$

Taking the curl of the Navier–Stokes equations eliminates the pressure and yields the vorticity equation. Using the vector identity

$$(u \cdot \nabla)u = \frac{1}{2} \nabla |u|^2 - u \times \omega,$$

the curl of the nonlinear term becomes $-\nabla \times (u \times \omega)$, and the pressure term ∇p vanishes since $\nabla \times (\nabla p) = 0$.

The result is:

$$\partial_t \omega + u \cdot \nabla \omega = \omega \cdot \nabla u + \nu \Delta \omega.$$

The term $\omega \cdot \nabla u$ is the **vortex stretching term**. It represents the amplification of vorticity by the velocity gradient along the direction of the vortex lines.

9.1. The Biot–Savart law

Given the vorticity ω , the velocity can be recovered via the Biot–Savart law:

$$u(x, t) = \frac{1}{4\pi} \int \frac{(x - y) \times \omega(y, t)}{|x - y|^3} dy.$$

⁴V. Scheffer, “Hausdorff measure and the Navier-Stokes equations,” **Comm. Math. Phys.** 55, 97–112 (1977). DOI Zbl

⁵L. Caffarelli, R. Kohn, and L. Nirenberg, “Partial regularity of suitable weak solutions of the Navier-Stokes equations,” **Comm. Pure Appl. Math.** 35, 771–831 (1982). DOI Zbl

This expresses u as a singular integral of ω . In particular, ∇u is a singular integral operator applied to ω , so the vortex stretching term has the formal structure

$$\omega \cdot \nabla u \approx T(\omega) \cdot \omega,$$

where T is a Calderón–Zygmund operator. This quadratic, nonlocal interaction is the core of the difficulty: the velocity gradient that stretches the vorticity is itself determined by the vorticity through a singular integral.

9.2. The 2D vorticity equation

In 2D, the vorticity $\omega = \partial_1 u_2 - \partial_2 u_1$ is a scalar, and the vortex stretching term vanishes identically because ω is orthogonal to the plane of the flow. The equation reduces to

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega.$$

Thus vorticity is simply transported and diffused—a passive scalar advected by the velocity field. This is a major reason why 2D is tractable.

9.3. The 3D vorticity equation

In 3D, the vortex stretching term $\omega \cdot \nabla u$ is active and can amplify vorticity. If one writes $\alpha = \frac{\omega}{|\omega|}$ for the vorticity direction and $|\omega|$ for its magnitude, then

$$(\partial_t + u \cdot \nabla - \nu \Delta)|\omega| = (\alpha \cdot \nabla)u \cdot \alpha |\omega| + \nu |\nabla \alpha|^2 |\omega|.$$

The first term on the right represents stretching: the rate-of-strain matrix ∇u (symmetrized) acts on the direction field, producing growth proportional to $|\omega|$ itself. This is one of the main mechanisms suspected in any possible singularity formation.

So:

- in 2D, vorticity is transported;
- in 3D, vorticity can stretch and grow.

10. Why energy estimates are not enough

The kinetic energy $\frac{1}{2}\|u\|_{L^2}^2$ is the most basic dissipated quantity. However, energy is **supercritical** with respect to the natural scaling of the equations.

To see this, observe that if $u(x, t)$ solves Navier–Stokes with viscosity ν , then the rescaled field

$$u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$$

also solves the same equations (with the same ν). Under this scaling, the L^2 norm transforms as

$$\|u_\lambda(\cdot, t)\|_{L^2} = \lambda^{1-\frac{3}{2}} \|u(\cdot, \lambda^2 t)\|_{L^2} = \lambda^{-\frac{1}{2}} \|u(\cdot, \lambda^2 t)\|_{L^2}.$$

Thus the L^2 norm decays as $\lambda \rightarrow \infty$ (zooming in), meaning energy scale is weaker than the critical scaling. Norms that are **critical** (invariant under the scaling) include $\dot{H}^{\frac{1}{2}}$, L^3 , and BMO^{-1} .

Because the energy norm is supercritical, energy estimates alone do not control the norms relevant to singularity formation. To illustrate, the energy inequality provides a bound on ∇u in $L_t^2 L_x^2$, but the nonlinear term $(u \cdot \nabla)u$ requires control of u in L^4 or ∇u in L^2 in three dimensions—estimates that do not follow from energy bounds alone.

Hence any proof of global regularity must use something beyond soft energy arguments. One needs a deeper structural understanding of the nonlinearity.

This is reinforced by model equations and modified systems that preserve many of the same soft features, yet can exhibit blowup.

11. What was known classically

Before modern developments, one broadly knew:

- Leray–Hopf weak solutions exist globally.
- Smooth solutions exist locally in time.
- If singularities occur, they are very sparse in spacetime.
- 2D Navier–Stokes is globally well-posed.
- 3D global smoothness remained open.

The local well-posedness theory (Fujita–Kato, 1964) guarantees existence and uniqueness of smooth solutions on a short time interval $[0, T)$ for sufficiently regular initial data. In particular, the space $\dot{H}^{\frac{1}{2}}(\mathbb{R}^3)$ is critical for Navier–Stokes: for $u_0 \in \dot{H}^{\frac{1}{2}}$, there exists $T = T(\|u_0\|_{\dot{H}^{\frac{1}{2}}}) > 0$ and a unique mild solution

$$u \in C([0, T); \dot{H}^{\frac{1}{2}}) \cap L^2((0, T); \dot{H}^{\frac{3}{2}})$$

given by the Duhamel formula

$$u(t) = e^{\nu t \Delta} u_0 - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P}(u(s) \cdot \nabla) u(s) ds,$$

where $\mathbb{P}$ is the Leray projection onto divergence-free fields. The solution is smooth for $t \in (0, T)$. For more regular data $u_0 \in H^s$ with $s > \frac{1}{2}$, the solution belongs to $C([0, T); H^s) \cap L^2((0, T); H^{s+1})$.

The maximal time of existence $T_{\max}$ is characterized by the blowup criterion: if $T_{\max} < \infty$, then

$$\int_0^{T_{\max}} \|\nabla u(\cdot, t)\|_{L^\infty} dt = \infty$$

(or equivalently, the BKM criterion $\int_0^{T_{\max}} \|\omega\|_{L^\infty} dt = \infty$).

The prevailing community intuition for a long time leaned toward global regularity, though without proof.

12. Regularity criteria

A common strategy is: if one cannot prove global regularity directly, perhaps one can identify conditions under which a weak solution must be smooth.

12.1. Prodi–Serrin–Ladyzhenskaya criteria

A Leray–Hopf weak solution is smooth if it additionally belongs to a critical mixed-norm space. Specifically, if

$$u \in L_t^p L_x^q$$

with

$$\frac{2}{p} + \frac{3}{q} \leq 1 \quad q > 3$$

then the solution is smooth on $(0, T]$. The condition $\frac{2}{p} + \frac{3}{q} = 1$ defines the critical line; the strict inequality $q > 3$ excludes the endpoint.

The proof for $q < 6$ uses the Gagliardo–Nirenberg inequality

$$\|u\|_{L^q} \leq C \|u\|_{L^2}^{1-\theta} \|\nabla u\|_{L^2}^\theta,$$

where $\theta = 3\left(\frac{1}{2} - \frac{1}{q}\right)$. Combined with the Prodi–Serrin condition $\frac{2}{p} + \frac{3}{q} = 1$, one obtains enough control of the nonlinear term $|(u \cdot \nabla)u|$ to close the estimates via Gronwall’s inequality.

The endpoint case $q = 3$ (i.e., $p = \infty$) is critical and was substantially harder.

12.2. Escauriaza–Seregin–Šverák

Escauriaza, Seregin, and Šverák (2003) proved the endpoint $q = 3$ case: if

$$u \in L_t^\infty L_x^3$$

then the solution is smooth.⁶ The proof uses a backward uniqueness argument for the vorticity equation: assuming blowup, one shows that the vorticity must vanish identically, leading to a contradiction.

These regularity criteria are useful because they imply that any blowup must violate them in a specific quantitative way.

12.3. Beale–Kato–Majda criterion

For Euler, and in related forms for Navier–Stokes, a basic blowup criterion says that if a smooth solution blows up at time T , then

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty} dt = \infty.$$

7

So finite-time blowup requires vorticity to become sufficiently large.

12.4. Constantin–Fefferman criterion

Constantin and Fefferman (1993) refined this by showing that not only magnitude but also the geometry of vorticity matters.⁸ Define the vorticity direction $\xi = \frac{\omega}{|\omega|}$ wherever $\omega \neq 0$. If

$$\int_0^T \|\nabla \xi(\cdot, t)\|_{L^\infty}^2 dt < \infty$$

then the solution remains smooth up to time T . In other words, a singularity can form only if the direction field of the vorticity develops sufficiently rapid spatial oscillations. This geometric condition complements the BKM magnitude criterion and suggests that blowup, if it occurs, must be highly structured.

13. Sharp well-posedness spaces

Researchers then sought the largest function spaces in which one can still prove well-posedness.

⁶L. Escauriaza, G. Seregin, and V. Šverák, “ $L_{3,\infty}$ -solutions of Navier-Stokes equations and backward uniqueness,” *Russ. Math. Surveys* 58, 211–250 (2003). DOI Zbl

⁷J. T. Beale, T. Kato, and A. Majda, “Remarks on the breakdown of smooth solutions for the 3-D Euler equations,” *Comm. Math. Phys.* 94, 61–66 (1984). DOI Zbl

⁸P. Constantin and C. Fefferman, “Direction of vorticity and the problem of global regularity for the Navier-Stokes equations,” *Indiana Univ. Math. J.* 42, 775–789 (1993). DOI Zbl

13.1. Koch–Tataru

Koch and Tataru (2001) proved global well-posedness for small initial data in the critical space BMO^{-1} : there exists $\varepsilon > 0$ such that for any divergence-free u_0 with $\|u_0\|_{\text{BMO}^{-1}} < \varepsilon$, there exists a unique global mild solution

$$u \in L_t^\infty \text{BMO}^{-1} \cap L_t^2 C^{0, \frac{1}{2}}$$

given by the fixed point of the integral equation

$$u(t) = e^{\nu t \Delta} u_0 - B(u, u)(t),$$

where $B(u, v)(t) = \int_0^t e^{\nu(t-s)\Delta} \mathbb{P}(u(s) \cdot \nabla) v(s) ds$

and $\mathbb{P}$ is the Leray projection. The solution satisfies $t^{\frac{1}{2}} \|u(\cdot, t)\|_{L^\infty} \leq C \|u_0\|_{\text{BMO}^{-1}}$ and is smooth for $t > 0$.⁹

The space BMO^{-1} is the dual of the Hardy space H^1 and is essentially optimal: it is invariant under the natural scaling of the equations, and no larger scaling-critical space can be well-posed by the Bourgain–Pavlović result below.

13.2. Bourgain–Pavlović

Bourgain and Pavlović (2008) showed that the border case $\dot{B}_{\infty}^{-1, \infty}$ (a slightly larger Besov space than BMO^{-1}) is ill-posed: for any $\delta, \varepsilon > 0$, there exists smooth initial data u_0 with $\|u_0\|_{\dot{B}_{\infty}^{-1, \infty}} < \delta$ but the corresponding solution u satisfies $\|u(\varepsilon)\|_{\dot{B}_{\infty}^{-1, \infty}} > \frac{1}{\delta}$, i.e., arbitrarily small initial data produce arbitrarily large solutions in arbitrarily short time. This **norm inflation** implies that the solution map is discontinuous at the origin in $\dot{B}_{\infty}^{-1, \infty}$.¹⁰

The construction uses initial data concentrated at a very high frequency N : the solution of the truncated system is computed explicitly, and the quadratic interaction transfers energy to even higher frequencies N^2 , producing the norm inflation before the viscous term can act.

Thus BMO^{-1} is essentially the largest space for which a well-posedness theory exists.

13.3. Germain–Pavlović–Staffilani

Germain, Pavlović, and Staffilani (2007) showed that the Koch–Tataru solution enjoys higher regularity: it is real analytic in space for every $t > 0$, with uniform decay estimates

$$\|\partial^\alpha u(\cdot, t)\|_{L^\infty} \leq C_\alpha t^{-\frac{|\alpha|+1}{2}}$$

for any multi-index α .¹¹

The proof uses the mild formulation and shows that the solution can be expressed as a convergent power series in u_0 using the heat semigroup, with uniform estimates on each term. This implies, in particular, that solutions starting from small BMO^{-1} data become classical (in fact, analytic) instantaneously. As an application, they also proved that any self-similar solution in BMO^{-1} is smooth, complementing the Nečas–Růžička–Šverák nonexistence result.

So the picture is:

- at or below the critical threshold, one has good control;

⁹H. Koch and D. Tataru, “Well-posedness for the Navier-Stokes equations,” *Adv. Math.* 157, 22–35 (2001). DOI Zbl

¹⁰J. Bourgain and N. Pavlović, “Ill-posedness of the Navier-Stokes equations in a critical space in 3D,” *J. Funct. Anal.* 255, 2233–2247 (2008). DOI arXiv Zbl

¹¹P. Germain, N. Pavlović, and G. Staffilani, “Regularity of solutions to the Navier-Stokes equations evolving from small data in BMO^{-1} ,” *Int. Math. Res. Not.* 2007, rnm087 (2007). DOI arXiv Zbl

- above it, the global theory becomes much more delicate.

14. Numerical blowup searches

Given the difficulty of the theoretical problem, it is natural to search numerically for singularity candidates.

14.1. Kerr's anti-parallel vortex tubes

In 1993, Kerr numerically studied two anti-parallel vortex tubes with perturbed axes and observed rapid growth of maximum vorticity, consistent with a finite-time singularity. The Beale–Kato–Majda integral appeared to diverge, suggesting blowup.¹²

Later, Hou and Li revisited the same configuration with an adaptive mesh refinement method using much higher resolution and found no evidence of blowup. The apparent singular growth in Kerr's simulation was an artifact of underresolution near the vortex core: the vorticity growth saturated at later times and followed an algebraic, rather than singular, scaling.¹³

This illustrates a central challenge:

- quantities may grow rapidly and seem to diverge;
- but at later times they may deplete instead.

14.2. Other candidate scenarios

More complicated constructions, such as folded vortex sheets, interacting vortex rings, and multiscale structures, have also been proposed as possible blowup mechanisms. For axisymmetric Euler with swirl, numerical studies by Grauer and Sideris, as well as by Cichocki, showed transient growth but no conclusive singularity. The equations repeatedly resist definitive numerical confirmation.

14.3. Core numerical difficulties

Numerically detecting blowup is intrinsically hard because:

1. if a true singularity is approached, resolution requirements become extreme;
2. truncation and discretization errors can dominate;
3. artificial numerical viscosity may suppress the effect one is trying to detect;
4. numerical evidence alone cannot establish a theorem.

15. Computer-assisted proofs

A more rigorous numerical paradigm is the **computer-assisted proof**.

15.1. General strategy

The idea is:

1. Compute a highly accurate approximate solution $| (u)$.
2. Formulate the PDE as a zero-finding problem $F(u) = 0$ for a nonlinear operator F on a Banach space.
3. Apply the **Newton–Kantorovich theorem**: if F is Fréchet-differentiable, $F(|(u))$ is small, and $F'(|(u))$ is invertible with controlled norm, then a true solution u^* exists near $| (u)$.
4. Use **interval arithmetic** to enclose all floating-point computations in rigorous bounds, so that the error bounds on $F(|(u))$ and $F'(|(u))^{-1}$ are mathematically guaranteed.

¹²R. M. Kerr, "Evidence for a singularity of the three-dimensional, incompressible Euler equations," **Phys. Fluids A** 5, 1725–1746 (1993). DOI

¹³T. Y. Hou and R. Li, "Dynamic depletion of vortex stretching and non-blowup of the 3-D incompressible Euler equations," **J. Nonlinear Sci.** 16, 639–664 (2006). DOI

Thus one upgrades a numerical candidate into a theorem.

15.2. Interval arithmetic

Instead of floating-point numbers, one propagates intervals guaranteed to contain the true values. This provides rigorous error bounds.

This methodology has become increasingly effective in PDE over the last decade, thanks to advances in software, hardware, and analysis.

16. Tao's perspective on the difficulty

Terence Tao emphasized that any proof of global regularity must go beyond standard energy methods.

16.1. Logarithmically supercritical models

For the hyperdissipative Navier–Stokes equations, the Laplacian $\nu\Delta$ is replaced by $\nu(-\Delta)^\alpha$ with $\alpha > 1$. The critical threshold is $\alpha = \frac{5}{4}$ in three dimensions: for $\alpha > \frac{5}{4}$ (subcritical) one can prove global regularity, while standard Navier–Stokes corresponds to $\alpha = 1$ (supercritical), for which the problem remains open.¹⁴

Tao improved this by proving global regularity even for the borderline case $\alpha = \frac{5}{4}$, provided the dissipation is strengthened by a logarithmic factor: he considered a Fourier multiplier D with symbol $m(\xi) = |\xi|^{\frac{5}{4}} \frac{1}{\log(2 + |\xi|^2)^{\frac{1}{4}}}$, for which the dissipation is just marginally stronger than the critical case, yet global regularity holds.

This shows that the problem is delicately balanced: a small increase in dissipation at the critical scaling changes the answer dramatically.

16.2. Averaged Navier–Stokes and blowup models

Tao also constructed modified or averaged Navier–Stokes-type systems that preserve many soft features of the original equations—energy identity, Sobolev estimates, symmetries—yet admit finite-time blowup. In these models, the nonlinear term $(u \cdot \nabla)u$ is replaced by an averaged or truncated version that retains the same energy estimates and scaling properties but removes certain cancellations present in the genuine nonlinearity. The blowup is constructed via a self-similar ansatz combined with an Ornstein–Uhlenbeck-type stochastic construction.

Conclusion: soft properties alone cannot distinguish global regularity from blowup. One must use the precise structure of the genuine Navier–Stokes nonlinearity.

17. Onsager's conjecture and convex integration

The talk then turned to a different but related line of development: very rough solutions to Euler and Navier–Stokes.

17.1. Onsager's conjecture

For weak solutions of 3D Euler with Hölder continuity C^α (or, more generally, Besov regularity $B_{3,\infty}^\alpha$):

- if $\alpha > \frac{1}{3}$, energy is conserved;
- if $\alpha < \frac{1}{3}$, anomalous dissipation may occur, i.e., the time derivative $\frac{d}{dt} \int (\frac{1}{2})|u|^2 dx$ may be nonzero.

¹⁴T. Tao, “Global regularity for a logarithmically supercritical hyperdissipative Navier–Stokes equation,” *Anal. PDE* 2, 361–366 (2009). DOI arXiv

The conservation part was proved by Constantin, E, and Titi: if $u \in L^3((0, T); B_{3,\infty}^\alpha)$ with $\alpha > \frac{1}{3}$, then the kinetic energy $\frac{1}{2}\|u(t)\|_{L^2}^2$ is constant in time.¹⁵

17.2. Convex integration

The flexibility part was developed through the method of **convex integration**, introduced into fluid dynamics by De Lellis and Székelyhidi, drawing on ideas of Nash and Gromov.

A sequence of works improved the regularity threshold, culminating in Isett's (2018) proof of Onsager's conjecture: for every $\alpha < \frac{1}{3}$, there exists a nonzero weak solution $u \in C_t C_x^\alpha$ of the 3D Euler equations that has compact support in time and therefore fails to conserve energy.¹⁶

The proof combines the method of convex integration with a **gluing approximation** technique using **Mikado flows** (introduced by Daneri and Székelyhidi). Later, Buckmaster, De Lellis, Székelyhidi, and Vicol strengthened this: for any prescribed nonnegative energy profile $e(t)$, there exists a weak solution of the 3D Euler equations whose kinetic energy equals $e(t)$ for almost every t .

Thus very rough Euler solutions can behave wildly.

18. Nonuniqueness for Navier–Stokes

A striking modern development is that rough weak solutions to 3D Navier–Stokes can be nonunique.

18.1. Buckmaster–Vicol

Buckmaster and Vicol (2019) proved nonuniqueness for weak solutions of 3D Navier–Stokes: there exist at least two distinct global weak solutions $u, v \in L_t^\infty L_x^2 \cap L_t^2 \dot{H}^1$ with the same finite-energy initial data $u_0 \in L^2$. Moreover, these solutions can be chosen to be Hölder continuous with exponent $\alpha < \frac{1}{3}$ and to satisfy the energy equality (not just the inequality).¹⁷

The construction uses convex integration with **intermittent Beltrami flows**—a refinement of the Mikado flows used for Euler that incorporates a third scale parameter to control the nonlinear term at the level of the Navier–Stokes equations. The intermittent Beltrami flows are spatially periodic, divergence-free, almost Beltrami ($\nabla \times v \approx \lambda v$), and have disjoint supports in frequency space. By superimposing many such flows at different scales, one cancels the time derivative and diffusion errors while preserving the nonlinear structure. The convex integration scheme yields solutions u that are $C^{\frac{1}{3}-\varepsilon}$ in space and time, reflecting the Onsager critical exponent $\frac{1}{3}$.

However, these solutions are too rough to be Leray–Hopf weak solutions (they do not satisfy the energy inequality). So this does not directly settle the classical uniqueness question for Leray–Hopf solutions.

18.2. Jia–Šverák program

Jia and Šverák (2014, 2015) proposed a program to prove nonuniqueness for Leray–Hopf solutions. They gave sufficient conditions for nonuniqueness in terms of the spectral properties of the linearized operator around a self-similar, scale-invariant solution of the Navier–Stokes equations. If the linearized operator has an unstable eigenvalue, one can construct a second solution lying on the unstable manifold, branching off from the self-similar background solution. The verification of these spectral conditions is in principle approachable by numerical simulation, since they involve only

¹⁵P. Constantin, W. E, and E. S. Titi, “Onsager’s conjecture on the energy conservation for solutions of Euler’s equation,” **Comm. Math. Phys.** 165, 207–209 (1994). DOI

¹⁶P. Isett, “A proof of Onsager’s conjecture,” **Ann. of Math.** 188, 871–963 (2018). DOI arXiv Zbl

¹⁷T. Buckmaster and V. Vicol, “Nonuniqueness of weak solutions to the Navier–Stokes equation,” **Ann. of Math.** 189, 101–144 (2019). DOI arXiv Zbl

smooth functions.¹⁸

18.3. Albritton–Brue–Colombo

Albritton, Brué, and Colombo (2022) realized the Jia–Šverák program and proved nonuniqueness for forced Leray–Hopf solutions: there exist two distinct Leray–Hopf weak solutions of 3D Navier–Stokes, both with zero initial velocity $u_0 = 0$ and driven by the same smooth body force f , that coincide at $t = 0$ but differ at later times. The construction uses a self-similar, compactly supported vortex ring as the background solution and demonstrates that it is unstable under the Navier–Stokes dynamics in similarity variables.¹⁹

So with forcing, nonuniqueness at the Leray–Hopf level is known. The solutions live precisely on the borderline of the known well-posedness theory.

18.4. Announcement by Hou–Wang–Yang

In 2025, Hou, Wang, and Yang announced a computer-assisted proof of nonuniqueness for unforced Leray–Hopf solutions, building on the Jia–Šverák program. They construct a self-similar Leray–Hopf solution and rigorously certify the existence of an unstable eigenpair of the linearized operator by combining high-precision numerical computation with a decomposition of the operator into a coercive part plus a compact perturbation approximated by a finite-rank operator. The result establishes the existence of a second solution—indeed, infinitely many distinct Leray–Hopf solutions—for the same smooth, compactly supported initial data with zero external force.²⁰

If fully confirmed, this would represent a major breakthrough: nonuniqueness at the Leray–Hopf level without external forcing.

19. Finite-time blowup for Euler with boundary

A landmark result concerns 3D incompressible Euler in a cylindrical domain.

19.1. Luo–Hou numerical scenario

In 2014, Luo and Hou numerically studied axisymmetric 3D Euler in a cylinder with no-flow boundary on the solid wall and periodic boundary along the axial direction. Using a hybrid 6th-order Galerkin and 6th-order finite difference method on dynamically adaptive meshes, they observed a (3×10^8) -fold increase in the maximum vorticity and predicted a singularity time $t_s \approx 0.0035056$. The numerical data were checked against the Beale–Kato–Majda, Constantin–Fefferman, and Deng–Hou–Yu blowup criteria.²¹

A local analysis near the singularity suggested a self-similar blowup in the meridian plane. The geometry of the boundary appeared to play an essential role by creating a strong shear layer that drives the vorticity amplification.

19.2. Chen–Hou theorem

In a series of papers beginning in 2023, Chen and Hou proved finite-time blowup for the 3D incompressible Euler equations in a cylindrical domain with no-penetration boundary. Specifically, there exists smooth, finite-energy initial data (u_0, p_0) such that the corresponding solution of the 3D

¹⁸H. Jia and V. Šverák, “Are the incompressible 3D Navier–Stokes equations locally ill-posed in the natural energy space?” *J. Funct. Anal.* 268, 3730–3766 (2015). DOI arXiv Zbl

¹⁹D. Albritton, E. Brué, and M. Colombo, “Non-uniqueness of Leray solutions of the forced Navier–Stokes equations,” *Ann. of Math.* 196, 415–455 (2022). DOI arXiv Zbl

²⁰T. Hou, Y. Wang, and C. Yang, “Nonuniqueness of Leray–Hopf solutions to the unforced incompressible 3D Navier–Stokes equation,” arXiv:2509.25116 (2025). arXiv

²¹G. Luo and T. Y. Hou, “Potentially singular solutions of the 3D axisymmetric Euler equations,” *PNAS* 111, 12968–12973 (2014). DOI arXiv Zbl

axisymmetric Euler equations develops a singularity in finite time. The velocity field remains $C^{1,\alpha}$ and has finite energy up to the singularity time.²²

The proof is computer-assisted and follows the dynamic rescaling framework developed by Elgindi. The key steps are: (1) construct an approximate self-similar profile numerically; (2) linearize the rescaled equations around this profile and prove spectral stability; (3) upgrade to nonlinear stability via a fixed-point argument with rigorous error bounds using interval arithmetic.

19.3. Why this does not solve the Clay problem

This is a major theorem, but it does not resolve the Clay problem because:

- it concerns Euler, not Navier–Stokes;
- it takes place in a bounded domain with boundary;
- the boundary is crucial to the blowup mechanism.

So it does not answer the whole-space or periodic 3D Navier–Stokes regularity question.

20. Other blowup scenarios and dimension as a parameter

The lecture also mentioned numerical studies by Hou suggesting possible blowup scenarios for Navier–Stokes, including tornado-type structures with large vorticity amplification near a solid boundary.

Another interesting idea is to treat the spatial dimension d as a continuous parameter by defining a family of PDEs that interpolate between 2D (globally regular) and 3D (open). This is achieved by replacing the Biot–Savart kernel $|x - y|^{-(d-1)}$ with an analytic continuation in d . Numerically, one finds self-similar singularities for dimensions d above a critical threshold $d_c \approx 3.188$, while the equations remain regular for $d < d_c$. This suggests that 3D Navier–Stokes lies just below the critical dimension, and that one missing cancellation—such as a subtle depletion of the nonlinear term—might separate global regularity from blowup.²³

This is suggestive, not definitive, but it offers a parametric explanation for why the 3D problem is so delicately balanced.

21. AI and neural networks in singularity discovery

The lecture then discussed neural-network-based discovery methods.

21.1. Neural networks as nonlinear ansätze

A neural network represents a function by composing affine maps with nonlinear activation functions, for example ReLU. Because this is a highly nonlinear parametrization, such networks can approximate complicated structures with relatively few parameters.

21.2. Physics-informed neural networks

In a physics-informed neural network (PINN), one minimizes a loss function based on the PDE residual and possibly its derivatives or constraints.

Thus the PDE itself guides the search for candidate solutions.

21.3. Applications to singularity discovery

The speaker described joint work using neural-network-based methods to discover self-similar blowup profiles for equations related to fluid dynamics, such as:

²²J. Chen and T. Y. Hou, “Finite time blowup of 2D Boussinesq and 3D Euler equations with $C^{1,\alpha}$ velocity and boundary,” **Ann. PDE** 9, 14 (2023). arXiv:2208.14798; see also J. Chen and T. Y. Hou, **PNAS** 122, e2500940122 (2025). DOI: 10.1073/pnas.2500940122

²³T. Y. Hou, “Nearly self-similar blowup of generalized axisymmetric Navier–Stokes equations,” arXiv:2405.10916 (2024). arXiv

- **axisymmetric 3D Euler models:** the De Gregorio model and related 1D models that capture the vortex stretching mechanism;
- the **CCF equation** (Constantin–Cordoba–Fontelos), a toy model for the 3D Euler equations that retains the nonlocal quadratic structure but is simpler to analyze;
- **incompressible porous media (IPM) equations**, which model fluid flow through porous media and share many analytical features with the 2D surface quasi-geostrophic (SQG) equation.

For each equation, the neural network discovers candidate self-similar profiles by minimizing the PDE residual in the rescaled variables. The candidate can then be refined and, in some cases, rigorously verified using a Newton–Kantorovich argument.

These methods are good at discovering candidate singularity profiles, but they do not by themselves prove anything about Navier–Stokes.

22. Self-similar singularities

A major conceptual motif is the search for self-similar blowup.

22.1. Self-similar ansatz

Suppose blowup occurs at time T . The simplest ansatz is a self-similar solution of the form

$$u(x, t) = \frac{1}{(T - t)^\beta} U\left(\frac{x - x_0}{(T - t)^\gamma}\right),$$

or in a more general time-dependent rescaled form where the exponents are allowed to vary. For the Navier–Stokes equations, dimensional analysis (balancing $\partial_t u$ with $\nu \Delta u$) forces

$$2\beta + 1 = \gamma.$$

Here:

- β is the amplitude blowup exponent,
- γ is the spatial concentration exponent,
- U is the profile. The energy-critical case corresponds to $\beta = \frac{1}{2}$, $\gamma = 1$ (the Leray self-similar scaling), for which U is expected to belong to $L^3(\mathbb{R}^3)$.

Under the change of variables

$$\xi = \frac{x - x_0}{(T - t)^\gamma} \quad \tau = -\log(T - t),$$

the Navier–Stokes equations transform into

$$\partial_\tau U - \gamma \xi \cdot \nabla_\xi U + \beta U + (U \cdot \nabla_\xi) U = -\nabla_\xi P + \nu e^{-(2\beta+1-\gamma)\tau} \Delta_\xi U,$$

where $U = U(\xi, \tau)$ and P is the rescaled pressure. For the critical scaling $2\beta + 1 = \gamma$, the viscous term simplifies to $\nu \Delta_\xi U$, independent of τ , and the equation becomes autonomous in the similarity variables:

$$\partial_\tau U - \gamma \xi \cdot \nabla_\xi U + \beta U + (U \cdot \nabla_\xi) U = -\nabla_\xi P + \nu \Delta_\xi U.$$

A self-similar singularity corresponds to convergence to a stationary ($\partial_\tau = 0$) solution of this rescaled system as $\tau \rightarrow \infty$ (i.e., $t \rightarrow T^-$). The problem thus reduces to finding a nontrivial steady state $U_\infty(\xi)$ of the rescaled equations and proving that it is dynamically stable in an appropriate sense.

This is attractive because the complicated singular behavior in physical variables becomes a fixed-point problem in the rescaled variables.

22.2. Program for proving blowup

A possible route to a theorem is:

1. Find an approximate self-similar profile U_{app} numerically.
2. Linearize the rescaled equations around U_{app} :

$$\partial_\tau V = LV + N(V)$$

, where L is the linearized operator and N contains higher-order corrections.

3. Prove that L has a single unstable eigenvalue (corresponding to the instability of the profile) while the rest of the spectrum is stable, so the dynamics are governed by a one-dimensional unstable manifold.
4. Upgrade this to nonlinear control via a fixed-point argument, estimating the nonlinear terms in suitable weighted spaces.
5. Use interval arithmetic to make every constant rigorous.
6. Translate back to physical variables and conclude finite-time blowup.

22.3. Restrictions on self-similarity

Not every self-similar ansatz is admissible.

Caffarelli, in earlier lectures on the Millennium problems, emphasized that one cannot expect the simplest self-similar Navier–Stokes singularities directly, due to exclusion results such as those of Nečas, Růžička, and Šverák (1996). They proved that there are no nontrivial self-similar solutions of the form

$$u(x, t) = \frac{1}{(T - t)^{\frac{1}{2}}} U\left(\frac{x}{(T - t)^{\frac{1}{2}}}\right)$$

with $U \in L^3(\mathbb{R}^3)$ (the natural energy space for the profile). More precisely, if U is divergence-free and $U \in L^3(\mathbb{R}^3)$, then the only solution of this self-similar form is $U \equiv 0$.²⁴

This rules out the most naive self-similar blowup scenario for Navier–Stokes.

Hence, if one seeks singularity for Navier–Stokes, one may need first to understand Euler-type singularity formation and then show viscosity is lower order near the singularity.

Recent work of Constantin, Ignatova, and Vicol also places restrictions on possible self-similar exponents for Euler. For finite-energy data, the similarity exponent must satisfy $\gamma \geq \frac{2}{5}$ in general, and $\gamma \geq \frac{1}{2}$ for axisymmetric flows.²⁵ These constraints matter because they interact with the Navier–Stokes scaling.

23. Blowup from rough data for Euler

Another direction is to lower the regularity of the initial data.

23.1. Elgindi’s theorem

Elgindi (2021) proved finite-time blowup for 3D Euler in the whole space $\mathbb{R}^3$ (without boundaries). Starting from initial data $u_0 \in C^{1,\alpha}(\mathbb{R}^3)$ (Hölder continuous gradient) that is axisymmetric and has

²⁴J. Nečas, M. Růžička, and V. Šverák, “On Leray’s self-similar solutions of the Navier-Stokes equations,” *Acta Math.* 176, 283–294 (1996). DOI Zbl

²⁵P. Constantin, M. Ignatova, and V. Vicol, “On putative self-similarity for incompressible 3D Euler,” arXiv:2602.17570 (2026). arXiv

no swirl, the corresponding solution develops a singularity at a finite time $T < \infty$. The vorticity blows up like $\omega(t) \approx (T - t)^{-1}$ as $t \rightarrow T^-$, and the velocity gradient blows up like $\nabla u \approx (T - t)^{-1}$.²⁶

The proof uses a dynamic rescaling formulation: the solution is written in self-similar variables, and the problem is reduced to proving convergence to a stable, nontrivial stationary profile of the rescaled equations.

This was a major breakthrough: it showed singularity formation in the whole space without boundaries, though not from smooth data (the initial velocity is $C^{1,\alpha}$ but not C^2).

23.2. Stability and related works

Elgindi, Ghoul, and Masmoudi (2021) later proved that the blowup is **stable** under small perturbations within the class of axisymmetric no-swirl data. They constructed a neighborhood of the Elgindi initial data in the $C^{1,\alpha}$ topology such that every datum in this neighborhood also blows up in finite time, with the same self-similar rates. The proof combines a refined spectral analysis of the linearized operator in similarity variables with a nonlinear perturbation argument using weighted norms.²⁷

Other related works by Huang, Chen, Hou, and collaborators used multiscale constructions and analogous mechanisms to prove blowup for related models, including the 2D Boussinesq equations and the axisymmetric Euler equations with a boundary (see §19.2).

An important open question remains:

Can one upgrade these rough-data blowup constructions to smooth-data (C^∞) blowup for the 3D Euler equations in the whole space?

24. Where we stand

The landscape today is far richer than it was 25 years ago.

We now know:

- global existence of Leray–Hopf weak solutions;
- complete global theory in 2D;
- strong partial regularity theory;
- many conditional regularity criteria;
- nonuniqueness for very rough Navier–Stokes solutions;
- nonuniqueness for forced Leray–Hopf solutions;
- major announcements toward unforced Leray–Hopf nonuniqueness;
- finite-time blowup for 3D Euler in a cylinder;
- finite-time blowup for 3D Euler from rough initial data in the whole space;
- new numerical and AI-assisted tools for discovering candidate singularities.

But we still do not know:

- whether smooth 3D Navier–Stokes solutions can blow up in finite time;
- whether smooth solutions always exist globally;
- whether Leray–Hopf weak solutions are unique in the classical unforced setting.

²⁶T. M. Elgindi, “Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$,” *Ann. of Math.* 194, 647–727 (2021). DOI arXiv Zbl

²⁷T. M. Elgindi, T.-E. Ghoul, and N. Masmoudi, “On the stability of self-similar blow-up for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$,” *Camb. J. Math.* 9, 1035–1075 (2021). DOI arXiv

25. Outlook

According to the lecture, several broad future directions appear plausible:

1. Global regularity via new structure

Any proof must go beyond energy methods and exploit deep structure specific to Navier–Stokes.

2. Singularity formation through related models

Euler and modified equations may reveal the mechanisms needed for Navier–Stokes.

3. Leray–Hopf nonuniqueness and weak solution theory

New notions of solution, or refined understanding of old ones, may reshape the problem.

4. Computer-assisted proofs

These are becoming increasingly central.

5. AI-assisted discovery

While still exploratory, such methods may help identify candidate structures that can later be proved rigorously.

The central question remains:

Do there exist smooth initial data for 3D incompressible Navier–Stokes that produce a finite-time singularity, or are smooth solutions always global?

After more than 200 years of study, this remains unknown.

26. Clean mathematical summary

For reference, here is a concise statement of the main mathematical objects mentioned.

26.1. 3D incompressible Navier–Stokes

On $\mathbb{R}^3$ or $\mathbb{T}^3$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u + f$$

$$\nabla \cdot u = 0$$

$$u(\cdot, 0) = u_0.$$

26.2. 3D incompressible Euler

Set $\nu = 0$:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = f \quad \nabla \cdot u = 0.$$

26.3. Vorticity

$$\omega = \nabla \times u.$$

In 3D Navier–Stokes,

$$\partial_t \omega + u \cdot \nabla \omega = \omega \cdot \nabla u + \nu \Delta \omega.$$

In 2D Navier–Stokes,

$$\partial_t \omega + u \cdot \nabla \omega = \nu \Delta \omega.$$

26.4. Energy inequality for Leray–Hopf solutions

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 ds \leq \frac{1}{2}\|u_0\|_{L^2}^2.$$

26.5. Prodi–Serrin criterion

If

$$u \in L_t^p L_x^q \quad \frac{2}{p} + \frac{3}{q} \leq 1 \quad q > 3$$

then u is smooth.

26.6. Beale–Kato–Majda criterion

If a smooth solution blows up at time T , then

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty} dt = \infty.$$